

MECH 360 Notes

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1 Transformation of Stress

Given the 3-D stress tensor

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yy} & \sigma_{yx} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}, \quad (1.0.1)$$

the principal stress tensor is given by

$$\underline{\underline{\sigma_p}} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}. \quad (1.0.2)$$

Invariant Properties:

- $\text{trace}(\underline{\underline{\sigma}}) = \text{trace}(\underline{\underline{\sigma_p}})$
- $\det(\underline{\underline{\sigma}}) = \det(\underline{\underline{\sigma_p}})$

2 Pure Bending

2.1 Unsymmetric Bending Analysis

Main Ideas:

1. Moment is a vector - resolving vectors
2. "Principal" Centroidal axes - Mohr's circle
3. Superposition - adding stresses

This leads to showing how one unsymmetric bending problem is two symmetric bending problems.

Principal Centroidal Axes: Where the product of inertia is zero.

What is Product of Inertia?

Product of Inertia is a measure of body symmetry.

When you talk about symmetry, we talk about planes of symmetry.

If a body is symmetrical about a plane, let's say xy -plane, then

$$I_{xy} = 0$$

Recall,

$$I_{xy} = \int xy dA,$$

If

$$I_{zy} = 0, I_y \neq 0, I_z \neq 0,$$

then y and z are the principal centroidal axes (PCA). PCA's are mutually orthogonal. Finding principal planes is analogous to finding principal stresses on the Mohr's circle.

The area moment of inertia is a tensor too. We apply the Mohr's transformations to find PCA.

Any given section possess *principal centroidal axes* even if it is unsymmetric. Principal centroidal axes can be determined:

1. analytically
2. or using Mohr's circle.

If $\mathbf{M}$ is along the principal centroidal axis, the N.A. will be along the axis of $\mathbf{M}$, then the equations for symmetric members can be used to compute the stresses. The principal of superposition is used to determine stresses in general for unsymmetric cases.

Given some a couple moment $\mathbf{M}$, we have

$$M_z = M \cos \theta, \quad M_y = M \sin \theta,$$

then using superposition,

$$\sigma_x(y, z) = \frac{-M_z y}{I_z} + \frac{+M_y z}{I_y}.$$

Note, we are using a right-handed frame to achieve these signs.

Points along the N.A. have no stress, thus let $\sigma_x = 0$, and using $M_z = M \cos \theta$, $M_y = M \sin \theta$, we get

$$y = \underbrace{\left(\frac{I_z}{I_y} \tan \theta \right)}_a z,$$

representing a line $y(z)$ with slope a . Letting ϕ be the angle between the N.A. and the z -axis gives

$$\tan \phi = \frac{I_z}{I_y} \tan \theta.$$

“ Note, ϕ depends on the loading axis. Also,

$$\tan \phi = \frac{y_n}{z_n},$$

where (y_n, z_n) is some point on the NA.

Solving Steps: If $\underline{M}$ is not in the plane of symmetry, we have an unsymmetric bending problem.

1. Find PCA - (using Mohr)
2. Resolve moment vector about PCA
3. Find/apply $\sigma = \frac{-My}{I}$ for each component
4. Superposition - (assuming linearly elastic material)

Steps solving an example:

1. PCA: found by inspection
2. Decompose moment into cartesian components
3. Stress due to asymmetric bending is the sum of the stress due to each moment component

Example 8 Steps: For the given section, write the steps to find the bending stress and maximum bending stress.

1. Know/compute $\underline{I}$. If x - y are principal axes, $I_{xy} \neq 0$, which is an easy problem. To find I_{ij} where i and j represent x, y , or z , we find the centroid of each simple section, then apply the parallel axis theorem.
2. Draw Mohr's Circle to find principal moments of inertia and principal axes. In principal centroidal axis $\underline{I}_p = \begin{bmatrix} I_u & 0 \\ 0 & I_v \end{bmatrix}$. Note, $I_{uv} = 0$.
3. Superposition: $\sigma(u, v) = -\frac{M_v u}{I_v} + \frac{M_u v}{I_u}$ on an element at (u, v) . Since we can observe the direction of the moment and find a point $A : (u_A, v_A)$ that is farthest away from the neutral axis and causes each term in each direction to be of the same sign. Then, the max bending stress will be given by $|\sigma_{\max}| = \sigma(u_A, v_A)$. If unsure, you can try multiple points (u_i, v_i) and take $\max\{\sigma(u_i, v_i)\}$.

2.2 Eccentric Loading

Using superposition

$$\sigma_x(y, z) = \frac{P}{A} - \frac{M_z y}{I_z} + \frac{M_y z}{I_y}.$$

3 Shear Stresses in Beams and Thin-Walled Members

Consider the case when M not constant, $V = \frac{dM}{dx} \neq 0$. This is not pure bending since there is a varying bending moment.

- Shear stresses: $\tau = \frac{V}{A}$ is actually wrong
- Shear Stress Formula:

What induces shear stresses?

...

Most importantly: Always ask what is the shear plane? Visualize shear action, and calculate shear area.

3.1 Shear Flow

- q is shear flow
- V is shear force
- Q is first moment of area, units m^3 of some partial area of cross-section (dependent on question) about centroid

Shear flow:

$$q = \frac{VQ}{I} \quad [N/m]. \quad (3.1.1)$$

Moment of inertia for rectangular cross-section:

$$I = \frac{bh^3}{12}. \quad (3.1.2)$$

First moment of area:

$$Q = \int ydA \quad (3.1.3)$$

Note, Q changes with cut.

Then,

$$Q(y) = b(h/2 - y)\bar{y} \quad (3.1.4)$$

where

$$\bar{y} = (h/2 - y) \cdot 1/2 + y, \quad (3.1.5)$$

thus,

$$Q(y) = \frac{b}{2} \left(\frac{h}{2} - y \right)^2 + by \left(\frac{h}{2} - 2 \right) \quad [m^3], \quad (3.1.6)$$

which is a quadratic in y . $\implies q(y)$ varies quadratically in y .

*Importantly, equal and opposite shears exist at a cross-section.

Summary: Bending of thin-walled beams

- Shear flow is shear force per unit distance $q = \frac{VQ}{I}$
- q varies along the length of a beam (due to a change in V) as well as across the depth of cross-section (due to a change in $\sigma(y) = \frac{-My}{I}$)
- The sum total of shear flow at any given cross-section is conserved
- Shear force in a segment of cross-section: $F = \int qds$
- Shear flow can exhibit complex patterns in its direction and magnitude as illustrated next
- Most significantly shear flow can cause unwanted torsion during bending

Bending and Twisting

- Bending and Twisting: $\tau \neq \frac{VQ}{It}$
- Only Bending (No twisting) $\tau = \frac{VQ}{It}$. *Preferred for design.
- Only Twisting (No bending): $\tau \neq \frac{VQ}{It}$

3.2 Shear Center

I would like to emphasize that Q is for PART of the cross-sectional area.

- I is the moment of inertia calculated given by

$$\frac{bh^3}{12}$$

and Ad^2 .

For a beam in bending, there is shear flow.

I think we've just been ignoring it.

Steps for shear flow problem:

1. Find moment of Inertia of entire crosssection
2. Identify the area that you will find Q for. I think the area depends on what the nail of interest is in.
- 3.